Patch Based Detection of AI Generated Facial Images

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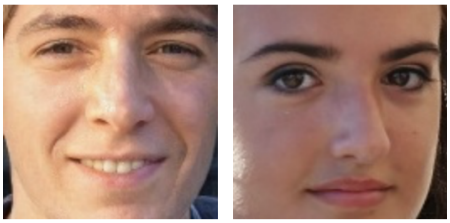
Introduction

In the current boom of artificial intelligence, there has been a significant increase in the production of AI-generated images and videos. As social media platforms become a primary source of news and information, they are increasingly saturated with synthetic media, much of which contains misleading or outright misinformation. The ability to prompt AI to create custom AI-videos and images has become increasingly accessible and uncomplicated. Yet the regulation of AI-generated content is not well defined across the internet and is increasingly harder to enforce. With this comes the widespread potential for abuse. The stakes are incredibly high for combatting scams, misinformation, and malicious content that utilize AI.



Fake 128x128 image on left, real image on right

AI-generated faces are a big part of AI being used for malicious purposes. Especially in cases where individuals attempt to impersonate others, disguise their identity, or deceive audiences. As a result, it is increasingly necessary to have tools to help detect these types of AI-images and videos. The nature of the AI field is ever-evolving and it is essential that the tools used to detect malicious content are also adapting. Because of this, it is important to allocate resources toward developing more accurate and efficient methods for identifying AI-generated faces.

In our research, our motivation was to create models that are able to effectively detect whether or not an image of a face is AI-generated or real. Our main research goal was to employ a patch-based analysis approach to detect AI-generated face images. In image classification, patch-based methods involve dividing each image into smaller sub-images (patches) sampled from the original image. For example a 128x128 pixel image can be split into four 64x64 patches or 16, 32x32 patches. Patch-based analysis reduces the size of each input image to the model while proportionally increasing the number of training samples available to the model. Increasing the training set size may come at the expense of decreasing global context for the smaller patch sized models.

128×128 full image on the left, with four 64×64 patches on the right

We aim to examine how patch size influences both classification accuracy and training time in patch-based neural network models for distinguishing AI-generated

from real facial images. Furthermore, with the intuition that patch sampling can expand limited training data more effectively than full-image approaches, we also investigate how training set size impacts convolutional neural network (CNN) accuracy, comparing patch-based models with full-image models for detecting AI-generated faces. Based on these results, we also aim to explore whether patch-based analysis in CNNs can offer greater interpretability for human understanding. To our knowledge, patch-based models and analysis have not yet been applied to detecting AI- generated facial images.

Background and Related Works:

The advancement of deep learning has caused the creation of highly realistic synthetic media, better known as "deep fakes." Generative Adversarial Networks(GANs) and other similar architectures are now able to produce photorealistic face images that are almost indistinguishable from authentic images to the naked eye. The patch-based detection approach stems from the nature of GAN artifacts; the differing signatures between real and synthetic images are often subtle and spatially localized instead of being uniformly distributed throughout the image. Specifically, the eyes, boundaries of hair follicles, and background transitions hold the most discriminative signal, while other regions usually appear almost identical to real images.

Deep CNN Architectures for Image Classification

Convolutional Neural Networks (CNNs), first demonstrated at scale by Krizhevsky, Sutskever, and Hinton (2012) through the landmark AlexNet architecture, established the foundational approach for image classification that this field continues to build on. By training a five-layer CNN with 60 million parameters on 1.2 million labeled images using GPU-accelerated computation, the authors achieved top-5 error rates of 17% on ImageNet, cutting the prior state-of-the-art nearly in half. Their innovations include utilizing ReLU activations for faster convergence, overlapping max-pooling, local response normalization, and dropout regularization, which became standard components of subsequent CNN architectures. This work established that depth and large-scale supervised training are the key ingredients for generalizable image representations, a principle that directly informs the CNN patch classifier design in this project.

Patch-Based Deepfake Detection

The Patch-based detection approach has been utilized before in "Patch-DFD: Patch-based end-to-end DeepFake discriminator" (Yu et al., 2022), in which Yu et al. argue that the standard approach of applying a backbone CNN to the entire face and feeding global features into a binary classifier is suboptimal, because the perceptual differences between real and fake faces tend to be subtle and spatially local. They introduce a Facial Patch Mapping strategy that extracts patches while preserving their original spatial information and spoof patterns, paired with a local voting scheme that aggregates patch-level predictions into a final image-level decision directly

paralleling the majority-vote aggregation used in this project. Crucially, they highlight that naively resizing patches before feeding them into a CNN can destroy the precise texture and frequency details that carry the forgery signal, underscoring the importance of preserving patch-native resolution. Their results demonstrate that this approach improves detection accuracy and training speed compared to global-feature baselines.

Methods

The dataset we utilized was from Kaggle which contained 140,000 256x256 RGB images: 70,000 AI generated facial images generated by Nvidia's StyleGAN and 70,000 real images of faces from Nvidia's Flickr image dataset. In the data cleaning process, we utilized python's face-recognition 1.3.0 library to crop images to focus on faces. This was driven by the goal of reducing noisy parts of images, particularly by removing possibly confounding features such as backgrounds. 422 Images were identified as flawed during this process and thus removed. For consistency, all images were then cropped on the centermost 128x128 pixels when being passed into our models or dataloaders.

The dataset was divided into training, validation, and test sets using an 80–10–10 split. The patch samples are taken after the split to prevent patches from the same image leaking between sets. Patch sampling occurs within PyTorch dataloaders, to dynamically allocate patches instead of downloading excessive amounts of images. The patch sizes utilized are 16x16, 32x32, 64x64, and 128x128 pixels. These specific patch sizes were chosen as they were powers of two and divided our 128x128 full images evenly. We trained 4 CNN on the full set of training data along with 1 naive neural network (the naive being the control). We then selected the best-performing patch-based model and compared it to the full 128×128 CNN using reduced training sets comprising 25% and 10% of the original training data. The test and validation sets in these reduced cases remained the same.

We utilized Python's PyTorch library for building our models. The model architectures we considered were convolutional neural networks and naive neural networks. The architecture for the convolutional neural network consisted of three convolutional layers (32, 64, & 128 filters) followed by two fully connected layers (64 & 2 as outputs). Max pooling was applied after the first and third convolution layers. Batch normalization was utilized after every convolutional layer. Dropout was used after the first fully connected layer to reduce overfitting. ReLu was used as the loss function. The naive neural network simply uses four fully connected layers (64, 32, 16, & 2 as outputs) with ReLu as the loss function. The architecture stayed consistent model to model—no matter patch size.

We trained our models using Google Collab's A100 gpu runtime. For all models, we used the adam optimizer (with .001 learning rate) and cross-entropy loss. Because the number of samples

varies because of patch sampling, batch sizes were adjusted so that each model performed the same number of steps per epoch—6,950 steps per epoch over the full training set. Each model was trained using early stopping based on validation loss. For the full training set, training was stopped when the validation loss failed to decrease from the observed minimum for three consecutive epochs, after which the model was reverted to and saved at its best-performing state. For the reduced datasets (25% and 10% of the training data), a patience of five epochs was used before reverting to the best model. Training, validation, and test accuracy were recorded for final models, along with test accuracy aggregated across patch samples for the patch-based CNNs. The aggregation accuracy predicts an image to be AI if at least half of the patches are predicted to be AI from the model. Training time and epochs to convergence were also recorded. These metrics exclude any training that occurred beyond the optimal model as determined by the use of validation loss early stopping.

Results:

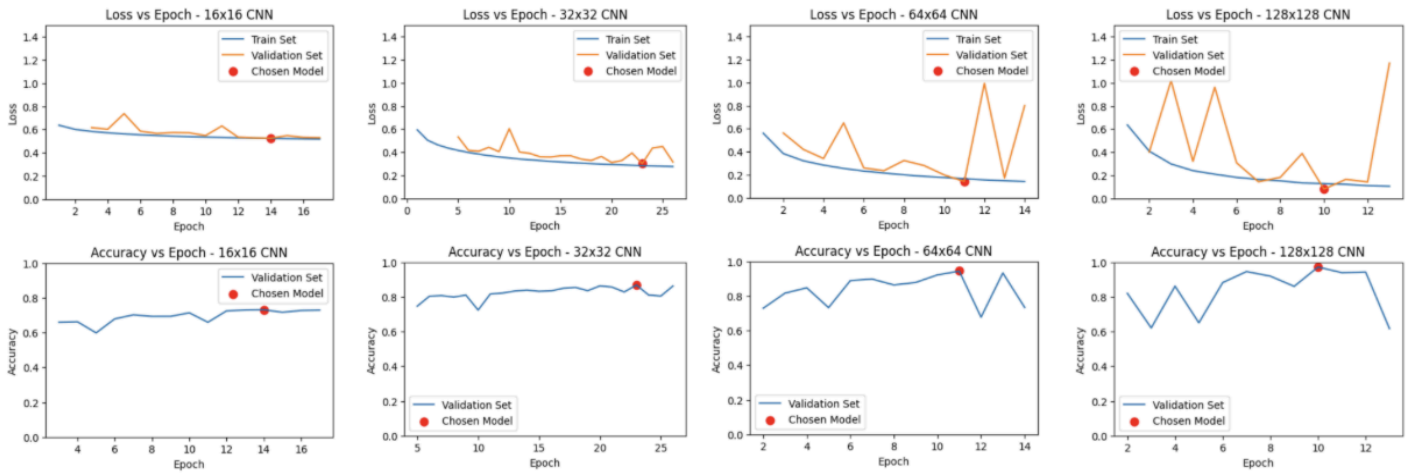
Full Training Set Models

Models	Final Train Accuracy	Final Validation Accuracy	Final Test Accuracy	Aggregated Accuracy	Time to Train	Epochs to convergence
128x128 CNN	97.48%	97.23%	97.10%	-	596.13 sec.	10
64x64 Patch CNN	94.89%	94.38%	94.34%	98.01%	676.66 sec.	11
32x32 Patch CNN	88.08%	86.88%	86.85%	99.17%	3366.77 sec.	23
32x32 Naive NN	Will	Add	Info	Here	Soon	!
16x16 Patch CNN	73.74%	73.18%	73.11%	91.78%	6852.52 sec.	14

The results from our five full models are summarized in the above table. The best performing model over the test set was the 32x32 patch-based CNN, which had an overall aggregated image accuracy of 99.17% and a single patch accuracy of 86.85%. The 64x64 patch-based CNN and the 128x128 CNN were the second and third most accurate: 98.01% and 97.10% respectively. Although the 32x32 patch-based CNN achieved approximately a 1% improvement in performance, it required significantly more training time—about 3366.77 seconds (56 minutes, 6 seconds) compared to roughly 600 seconds for both the 128x128 and 64x64 models.

The 16x16 patch-based CNN achieved a 91.78% aggregated test set image level accuracy and a 73.11% test set patch accuracy. Although this model took 6852.52 seconds of total training time (114.2 min). This was two times as long as the 32x32 patch-based CNN and over 10 times as long as the 64x64 patch-based CNN and 128x128 CNN.

WILL ADD INFO ABOUT THE 32x32 NAIVE NN HERE !!!!!



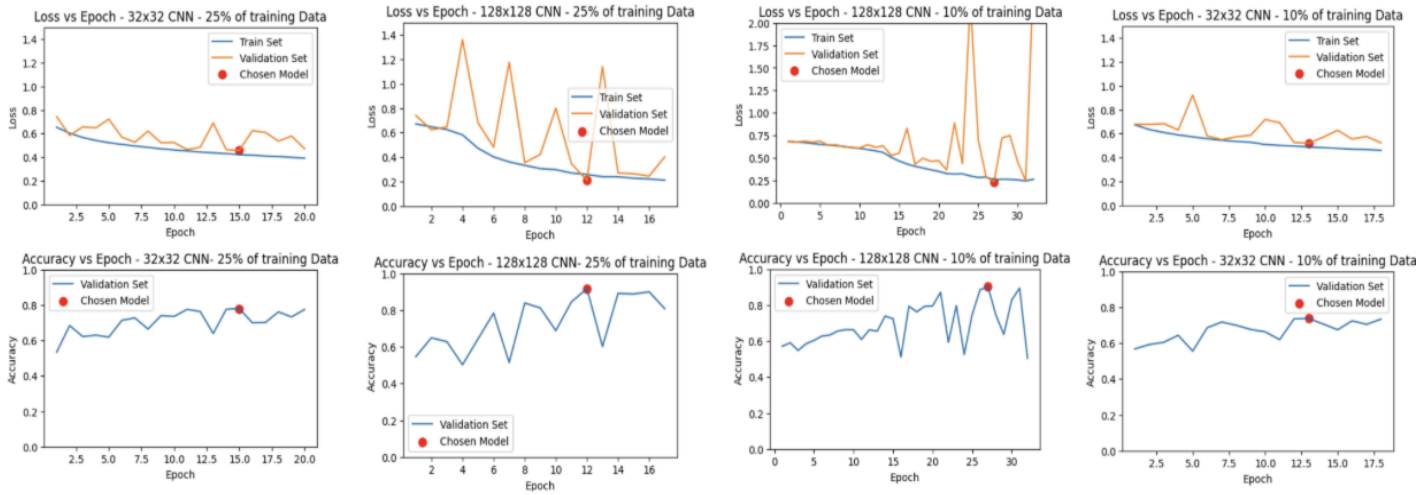
Training Set Restricted Models

Models	Final Train Accuracy	Final Validation Accuracy	Final Test Accuracy	Aggregated Accuracy	Time to Train	Epochs to convergence
128x128: 25% Train Set Size	92.23%	91.75%	91.95%	-	203.27 sec.	12
32x32 Patch CNN 25% Train Set Size	78.96%	77.95%	77.99%	90.35%	554.12 sec	15
128x128: 10% Train Set Size	91.21%	90.16%	90.37%	-	185.74 sec.	27
32x32 Patch CNN 10% Train Set Size	74.82%	73.74%	73.55%	86.02%	196.10 sec.	13

The above table has results for our four restricted training set models. The best performing model for both the 25% and 10% restrictions was the 128x128 CNN. Its test set accuracy was 91.95%

when trained on the 25% dataset and 90.37% when trained on the 10% dataset. The 32×32 patch-based CNN performed closely on the 25% training set, achieving an aggregated test accuracy of 90.35%. However, its performance decreased further on the 10% training set, dropping to 86.02%.

Training time decreased for all models compared to the full training set. The 32×32 patch-based CNN showed the largest reduction in training time, decreasing by a factor of 5 on the 25% training set and by a factor of 15 on the 10% training set. The 128x128 CNN decreased by a factor of 3 for both reduced training set models.



Discussion:

One of the most noteworthy findings of this project is the gap between patch-level and image-level accuracy. Individual 32×32 patches are classified at roughly 86.85% accuracy, yet aggregating those predictions via majority vote across all patches yields approximately 99.17% image-level accuracy on the test set. This result reflects a core property of ensemble-style classification: independent errors at the patch level are unlikely to be correlated across all patches of a given image, so the majority vote reliably corrects for individual patch misclassifications. Full-image models, which have access to the entire face in a single forward pass, also achieve strong accuracy (90%+ across all tested scales), and interestingly retain more accuracy than patch-based models when input resolution is reduced. This likely occurs because a full-image model preserves global structural context such as facial symmetry, inter-feature consistency, and background coherence, that is discarded when the image is divided into small patches. Patch-based models trade some of this contextual information for greater sensitivity to local artifacts.

A distinctive advantage of the patch-based approach is its interpretability. The heatmap visualization, in which each patch is colored red (predicted fake) or green (predicted real) with

intensity proportional to the model's confidence, displays spatial localization of suspected forgery artifacts that a binary whole-image classifier cannot offer. As illustrated in the sample outputs, the model frequently concentrates high-confidence fake predictions around facial boundaries, hair regions, and background transitions, which aligns with the known tendency of GAN generators to struggle with fine detail in these areas. This form of human-readable evidence is practically valuable in contexts where a binary label alone is insufficient, such as content moderation workflows, forensic investigations, or educational demonstrations of how AI-generated images differ from photographs. The patch-based approach also connects directly to the literature: Patch-DFD (Yu et al., 2022) similarly found that a local voting scheme over patch-level predictions improves both accuracy and the explainability of the detection decision.

Green indicates predictions labeled as real, while red indicates predictions labeled as AI-generated. Label 0 corresponds to real, and label 1 corresponds to AI.

Conclusion

The proliferation of AI-generated synthetic media represents a growing sociopolitical threat. As Vaccari and Chadwick (2020) demonstrate, even when people are not directly deceived by deepfakes, exposure to them erodes trust in authentic media, leading to broader civic cynicism. Robust detection tools are therefore a critical infrastructure for maintaining information integrity. This project presents a patch-based CNN approach to detecting AI-generated face images, achieving approximately 97.4% image-level accuracy on a balanced test set of 13,900 images through majority-vote aggregation of patch-level predictions. Across all tested configurations, full-image CNN models consistently achieved 90%+ accuracy, demonstrating that modern CNNs can reliably distinguish synthetic from authentic face images when trained on sufficiently large datasets.

These results come with important limitations. Each reported accuracy figure reflects a single training run rather than an average over multiple runs, meaning the statistics carry unknown variance. Given more time, repeated training with different random seeds and averaging of results would provide a more statistically reliable estimate of model performance. Additionally, all models were trained and evaluated on the same GAN architecture distribution; cross-generator generalization, meaning the ability to detect synthetic images from architectures not seen during training, including diffusion models, remains untested. Future work should prioritize these evaluation conditions. More broadly, this project reinforces the value of exploring unique model selection strategies in the detection literature. Patch-based detection, as validated both in this study and in prior work such as Patch-DFD (Yu et al., 2022), offers a compelling combination of competitive accuracy and spatial interpretability that whole-image classifiers cannot match. As synthetic media generation continues to improve, the development of interpretable, generalizable, and ensemble-based detection methods will be essential to staying ahead of the threat.

References:

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